

Primary sources & further reading

Part 05 — 🍌 Generative AI · 05-genai/papers/PAPERS.md · 05-genai/resources/RESOURCES.md

Papers & Primary Sources — Generative AI

`bash ../_tools/fetch_papers.sh 05-genai` downloads the arXiv PDFs below into this folder.

The canon (read in this order)

#	Paper	Year	Why it matters	Link
1	Denoising Diffusion Probabilistic Models (DDPM) — Ho, Jain, Abbeel	2020	The paper <code>code/diffusion_from_scratch.py</code> implements. Forward/reverse process, the noise-prediction objective, the derivation behind every equation in notes/01.	arXiv:2006.11239
2	Denoising Diffusion Implicit Models (DDIM) — Song, Meng, Ermon	2020	Deterministic, non-Markovian sampling — the reason production tools sample in 20-50 steps instead of 1000. <code>code/guidance_and_sampling.py</code> measures exactly this trade-off.	arXiv:2010.02502
3	Classifier-Free Diffusion Guidance — Ho, Salimans	2022	The w-scaled guidance equation every text-to-image tool exposes as "CFG scale" / "guidance scale." No separate classifier needed — just condition dropout during training.	arXiv:2207.12598
4	High-Resolution Image Synthesis with Latent Diffusion Models — Rombach, Blattmann, Lorenz, Esser, Ommer	2022	This paper is Stable Diffusion: VAE-compressed latent space + diffusion U-Net + cross-attention text conditioning.	arXiv:2112.10752
5	Scalable Diffusion Models with Transformers (DiT) — Peebles, Xie	2022	Replaces the U-Net backbone with a Vision-Transformer-style architecture; the backbone behind Stable Diffusion 3, Flux, and Sora.	arXiv:2212.09748
6	Learning Transferable Visual Models From Natural Language Supervision (CLIP) — Radford et al.	2021	Contrastive text-image alignment. Powers the original Stable Diffusion text encoder, the CLIP-score metric (notes/02 §5), and every LLaVA-style vision encoder. Also the retrieval backbone in Module 08.	arXiv:2103.00020
7	Visual Instruction Tuning (LLaVA) — Liu, Li, Wu, Lee	2023	The recipe notes/02 §4.1 walks through: frozen CLIP encoder + small projection + off-the-shelf LLM. The template nearly every open vision-language model still follows.	arXiv:2304.08485

Going deeper

Paper	Year	Takeaway	Link
Generative Adversarial Networks — Goodfellow et al.	2014	The original minimax generator-vs-discriminator paper. Read it to understand <i>why</i> mode collapse and training instability are structural, not implementation bugs.	arXiv:1406.2661
Auto-Encoding Variational Bayes (VAE) — Kingma, Welling	2013	The direct ancestor of latent diffusion's compression stage. Already in Module 02's papers [02-deep-learning/papers/PAPERS.md] — read it there if you haven't.	arXiv:1312.6114
Improved Denoising Diffusion Probabilistic Models — Nichol, Dhariwal	2021	The cosine noise schedule (lab exercise 2) and learned variance — the paper that took DDPM from "works" to "competitive."	arXiv:2102.09672
Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding (Imagen) — Saharia et al.	2022	Google's case for a large frozen T5 text encoder over CLIP — directly informs why notes/02 §1 says newer systems add an LLM/T5 encoder alongside CLIP.	arXiv:2205.11487
Video generation models as world simulators (Sora technical report) — OpenAI	2024	Spacetime-patch tokenization for video diffusion — the "extra axis on DiT" story in notes/02 §2. No peer-reviewed paper exists; this is the primary source.	openai.com/research/video-generation-models-as-world-simulators
Adding Conditional Control to Text-to-Image Diffusion Models (ControlNet) — Zhang, Rao, Agrawala	2023	Structural conditioning (edge/depth/pose) alongside a frozen base model — decouples <i>what</i> from <i>where</i> , the production unlock covered in notes/02 §1.	arXiv:2302.05543
Scaling Rectified Flow Transformers for High-Resolution Image Synthesis (SD3) — Esser et al.	2024	Flow matching as diffusion's generalization — what trained Stable Diffusion 3 and Flux; explains the "close cousin" hedge in notes/01 §1.	arXiv:2403.03206
Consistency Models — Song, Dhariwal, Chen, Sutskever	2023	Distill a diffusion model into a 1-4 step sampler — the frontier of attacking diffusion's one real weakness, sampling cost.	arXiv:2303.01469

How to read a diffusion paper (30 minutes, 3 passes)

- Pass 1 (5 min):** what's being diffused (pixels? a VAE latent? spacetime patches?), and what's the conditioning signal (text, class label, control map, nothing)?
- Pass 2 (15 min):** the sampler. Stochastic or deterministic? How many steps does the paper actually report needing, and against what baseline?
- Pass 3 (10 min):** the evaluation section. Which of FID / CLIP score / human preference did they use, and what does notes/02 §5 say that metric can't tell you? Papers rarely volunteer their metric's blind spot — you have to bring it yourself.

Keep a one-paragraph note per paper. In six months the note is worth more than the PDF.

Resources — Generative AI

Courses (free)

Course	Who it's for	Link
Hugging Face — Diffusion Models Course	The direct sequel to this module: hands-on diffusers from DDPM basics through Stable Diffusion internals	https://huggingface.co/learn/diffusion-course
Hugging Face — Computer Vision Course	Broader CV grounding, including generative sections	https://huggingface.co/learn/computer-vision-course
fast.ai — Practical Deep Learning, Part 2	Builds Stable Diffusion from its components, top-down, code-first	https://course.fast.ai/Lessons/part2.html
Stanford CS236 — Deep Generative Models	The rigorous treatment: VAEs, GANs, flows, diffusion, with the math	https://deepgenerativemodels.github.io/
MIT 6.S183 — Generative AI with Diffusion Models	Focused university course, lecture notes + labs	https://mit-6s183.github.io/
DeepLearning.AI — How Diffusion Models Work	Short, practical, by the Hugging Face diffusers team	https://www.deeplearning.ai/short-courses/how-diffusion-models-work/
DeepLearning.AI — Building Systems with the ChatGPT API / Multimodal RAG courses	Practical LLaVA-style multimodal pipelines	https://www.deeplearning.ai/short-courses/

Repositories worth cloning

Repo	What's inside
https://github.com/huggingface/diffusers	The production diffusion library — DDPM/DDIM/DPM-Solver schedulers, pipelines for SD/SDXL/SD3/Flux, ControlNet, all in readable Python
https://github.com/CompVis/stable-diffusion	The original Stable Diffusion (v1) research repo — read this to see notes/01's latent-diffusion equations as actual code
https://github.com/Stability-AI/stablediffusion	Stability AI's SD 2.x release repo
https://github.com/AUTOMATIC1111/stable-diffusion-webui	The most widely used community UI — samplers, ControlNet, LoRA, inpainting, all exposed as knobs
https://github.com/comfyanonymous/ComfyUI	Node-graph UI for diffusion pipelines — the best way to see the text-to-image pipeline diagram in notes/02 §1 as a literal graph you wire up
https://github.com/openai/CLIP	Reference CLIP implementation and pretrained checkpoints
https://github.com/haotian-liu/LLaVA	Reference LLaVA implementation — the recipe in notes/02 §4.1, in code
https://github.com/facebookresearch/DiT	Reference Diffusion Transformer implementation (Peebles & Xie)
https://github.com/lucidrains/denoising-diffusion-pytorch	A clean, minimal PyTorch DDPM/DDIM — the natural next step up from <code>code/diffusion_from_scratch.py</code>
https://github.com/facebookresearch/audiocraft	MusicGen and AudioGen — the audio generation family from notes/02 §3

Datasets and pretrained checkpoints

Resource	Why	Where
LAION / LAION-Aesthetics	The web-scale image-text dataset behind most open text-to-image models — read the licensing discussion in notes/02 §7 before assuming you can use it commercially	https://laion.ai/
Hugging Face Hub — models	Pretrained diffusion, CLIP, and VLM checkpoints, one line to load	https://huggingface.co/models
Papers with Code — Image Generation	Leaderboards with FID numbers so you can see the metric from notes/02 §5 in the wild	https://paperswithcode.com/task/image-generation

Tools

Tool	Use
Hugging Face <code>diffusers</code>	The library — build a pipeline in under 10 lines once you've done the labs here
ComfyUI / A1111 webui	Visual, no-code experimentation with samplers, guidance scale, ControlNet
<code>torchmetrics</code> (<code>FrechetInceptionDistance</code> , <code>CLIPScore</code>)	Compute the metrics notes/02 §5 critiques, so you can see their limits yourself
C2PA Content Credentials tools	Inspect/attach provenance metadata — https://c2pa.org/
Google SynthID	The most widely deployed invisible watermark — https://deepmind.google/technologies/synthid/

Communities

- Hugging Face forums (diffusion-models category) — <https://discuss.huggingface.co/>
- r/StableDiffusion — <https://www.reddit.com/r/StableDiffusion/>
- r/MachineLearning — <https://www.reddit.com/r/MachineLearning/>
- Papers with Code — <https://paperswithcode.com/>
- EleutherAI Discord (open generative-model research) — <https://www.eleuther.ai/>

Cheat sheets

- Hugging Face `diffusers` scheduler comparison — <https://huggingface.co/docs/diffusers/using-diffusers/schedulers>
- Also see `../../_shared/cheatsheets/`